

In-Class 2. Categorical Data Analysis

Nishimura/Suzer-Gurtekin

January 2024

Group Assignments for Today's In-class Exercises

Maryland1		Maryland2	
Kangrui Liu		Da'Sauni E Singleton	
Katherine Pistner		Lindsey Ann English	
Lucy Robles		Mathew Hill	
Matthew Cowles		Michael Joseph Muldoon	
Xiniao Zhao		William Hopkins	
Michigan1		Michigan2	
Avery Muhl		Ang Li	
Chloe Chen		Aria IU	
Keqing Lu		Qichuan Zhou	
Nia Holland		Qiyu Yang	
Sol Rabine		Wen Chen	
Michigan3		Michigan4	
Akari Oya		Peter Tan	
Leng Seong Che		Rona Hu	
Saujanya Acharya		Yuya Okubo	
Yesdi Christian Calvin		Zhuoer Wang	
Zeyu Lou			
Zoom			
Angelo Cozzubo			
Elizabeth Seravello			
Eric Ohemeng			
Francis Lavoe			
Joshua Turnbull			
Queen Alicia Sampson			
Sean Johnson			
Srijeeta Mitra			
Tavril Saint Jean			

- I will create breakout rooms for the zoom participants during the class

- Identify your group members and sit at close proximity
- Take time to introduce yourself if you don't know each other before
- Select a spokesperson for each exercise to summarize your discussion back to the class

Overview

1 [Analysis of Two-Way Tables](#)

2 [Odds Ratios and Relative Risk](#)

Two-Way Tables

Rows=I

Columns=J

Columns						
Rows	1	2	3	4	5	
1	n_{11}	n_{12}	n_{13}	n_{14}	n_{15}	n_{1+}
2	n_{21}	n_{22}	n_{23}	n_{24}	n_{25}	n_{2+}
3	n_{31}	n_{32}	n_{33}	n_{34}	n_{35}	n_{3+}
4	n_{41}	n_{42}	n_{43}	n_{44}	n_{45}	n_{4+}
	n_{+1}	n_{+2}	n_{+3}	n_{+4}	n_{+5}	n

Two-Way Tables

In this notation, each cell is n_{ij} where i is the *row* and j is the *column*.

The plus sign denotes marginal totals:

$$n_{i+} = \sum_{j=1}^J n_{ij}$$

$$n_{+j} = \sum_{i=1}^I n_{ij}$$

$$n = \sum_{i=1}^I n_{i+} = \sum_{j=1}^J n_{+j} = \sum_{i=1}^I \sum_{j=1}^J n_{ij}$$

2×2 Tables

Table			
Condition	Present	Absent	Row Total
Yes	n_{11}	n_{12}	n_{1+}
No	n_{21}	n_{22}	n_{2+}
Column Total	n_{+1}	n_{+2}	n

In-Class Problem 1

	Disease		Row Total
	Yes	No	
Male	10	40	50
Female	20	30	50
Column Total	30	70	100

What is n_{+1} ?

What is n_{2+} ?

What is $Pr(M, D)$?

What is $Pr(D|F)$?

In-Class Problem 1

	Disease		Row Total
	Yes	No	
Male	10	40	50
Female	20	30	50
Column Total	30	70	100

What is n_{+1} ? 30

What is N_{2+} ? 50

What is $Pr(M, D)$? 0.10

What is $Pr(D|F)$? 0.40

Two-Way Tables

If the two variables are unrelated, then any cell proportion is the product of the marginal proportions. Using the notation from last time:

$$\pi_{ij} = \pi_{i+} \pi_{+j}.$$

This gives us a method for creating **Expected** counts if we want to test for independence.

Another way to write it, using our notation:

$$e_{ij} = np_{i+}p_{+j} = \frac{n_{i+}n_{+j}}{n}$$

In-Class Problem 2

The FREQ Procedure

Table of gender by age

gender		age					
Frequency							
Expected							
Cell Chi-Square							
	0-29	30-39	40-49	50-59	>=60		Total
males	185 180.22 0.1269	207 209.78 0.0368	260 257.46 0.0252	180 178.31 0.016	71 77.237 0.5036		903
females	4 8.7814 2.6034	13 10.222 0.7551	10 12.545 0.5163	7 8.6885 0.3281	10 3.7635 10.335		44
Total	189	220	270	187	81		947

Statistics for Table of gender by age

Statistic	DF	Value	Prob
Chi-Square	4	15.2461	0.0042
Likelihood Ratio Chi-Square	4	12.6670	0.0130
Mantel-Haenszel Chi-Square	1	4.8961	0.0269
Phi Coefficient		0.1269	
Contingency Coefficient		0.1259	
Cramer's V		0.1269	

Sample Size = 947

In-Class Problem 2

Please use the table on the previous page and calculate a X^2 test of association by hand (or in a spreadsheet), that is, not using R.

In-Class Problem 2

Now let's look at the solution in R. The following code:

```
chisq.test(data.matrix(drunk))
```

Produces the following output:

```
Pearson's Chi-squared test
```

```
data: data.matrix(drunk)
```

```
X-squared = 15.2461, df = 4, p-value = 0.004217
```

```
Warning message:
```

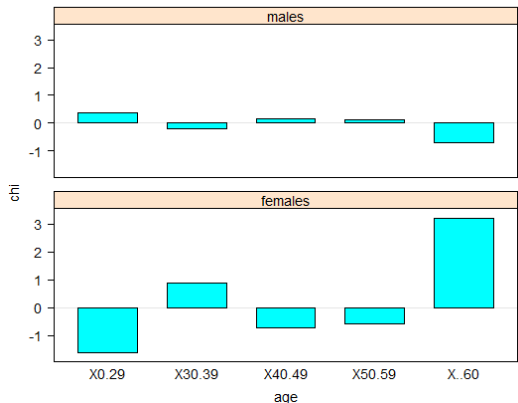
```
In chisq.test(data.matrix(drunk)) :
```

```
Chi-squared approximation may be incorrect
```

Example 1

We can also produce a graphic of the chi-square deviations:

chi deviations for drunk.dat



Example 2

Now let's look at the example in R. We have a *matrix* with our data:

```
> glasses
      delinq non.del
glasses      1      5
no.glasses   8      2
```

Example 2

We can request Fisher's exact test with the following code:

```
fisher.test(glasses)
```

Which produces the following output:

```
Fisher's Exact Test for Count Data
```

```
data:  glasses
p-value = 0.03497
alternative hypothesis: true odds ratio is not equal to
95 percent confidence interval:
 0.0009525702 0.9912282442
sample estimates:
odds ratio
0.06464255
```

OR and RR

	Disease	
	Yes	No
Male	0.15	0.35
Female	0.10	0.40

We want the conditional probability that you have disease given that you are male: $PR(D|M) = \pi_{1|1} = \frac{\pi_{11}}{\pi_{11} + \pi_{12}}$.

$$PR(D|M) = \pi_{1|1} = \frac{0.15}{0.15 + 0.35} = 0.3$$

Relative Risk

Now, define the relative risk.

$$\text{Relative Risk (Response Category 1)} = \frac{\pi_{1|1}}{\pi_{1|2}}$$

For example:

$$\frac{\pi_{1|1}}{\pi_{1|2}} = \frac{\Pr(D|M)}{\Pr(D|F)} = \frac{.3}{.2} = 1.5$$

Relative Risk

An easy way to estimate the relative risk is:

$$\frac{n_{11}/n_{1+}}{n_{21}/n_{2+}}$$

The distribution of the relative risk is highly skewed. Therefore, it is better to estimate the variance on the log scale.

$$\hat{V} \left\{ \ln \left(\frac{n_{11}/n_{1+}}{n_{21}/n_{2+}} \right) \right\} = \frac{1 - \frac{n_{11}}{n_{1+}}}{n_{11}} + \frac{1 - \frac{n_{21}}{n_{2+}}}{n_{21}} = \frac{\Pr(\bar{D}|M)}{n_{11}} + \frac{\Pr(\bar{D}|F)}{n_{21}}$$

This is the variance of the logarithm of the Relative Risk(Response Category 1).

In-Class Problem 3

	Disease	
	Yes	No
Male	20	40
Female	30	30

What is the relative risk for men relative to women?

What is the variance of this estimate? Leave it on the log scale, $\hat{V}(\ln(\hat{RR}))$.

In-Class Problem 3

	Disease	
	Yes	No
Male	20	40
Female	30	30

What is the relative risk for men relative to women? $\frac{20/60}{30/60} = \frac{.33}{.5} = 0.67$

What is the variance of this estimate? $\hat{V} \left(\ln(\hat{RR}) \right) = \frac{3}{60}$

Odds Ratio

Within row 1 the *odds* (not odds ratio) that the response is in column 1 instead of column 2 is defined as:

$$Odds_1 = \frac{\pi_{1|1}}{\pi_{2|1}} = \frac{\pi_{11}}{\pi_{12}}$$

From our example, this could be written:

$$\frac{\Pr(D|M)}{\Pr(\bar{D}|M)} = \frac{\Pr(D|M)}{1 - \Pr(D|M)}$$

Continuing the example, the odds that a man will have the disease are $\frac{.3}{1-.3} = .43$. For women, this odds are $\frac{.2}{1-.2} = .25$

Odds Ratio

Those are the odds. The ratio of the odds is called the *odds ratio*.

$$\theta = \frac{\pi_{1|1} / \pi_{2|1}}{\pi_{1|2} / \pi_{2|2}}$$

Remember that $\pi_{1|1} = \frac{\pi_{11}}{\pi_{11} + \pi_{12}}$. So we can also write it in the following manner:

$$\theta = \frac{\frac{\pi_{11}}{\pi_{11} + \pi_{12}} / \frac{\pi_{12}}{\pi_{11} + \pi_{12}}}{\frac{\pi_{21}}{\pi_{21} + \pi_{22}} / \frac{\pi_{22}}{\pi_{21} + \pi_{22}}} = \frac{\pi_{11} / \pi_{12}}{\pi_{21} / \pi_{22}} = \frac{\pi_{11} \pi_{22}}{\pi_{21} \pi_{12}}$$

Hence the name *cross-product ratio*.

Odds Ratio

We estimate the odds ratio using:

$$\hat{\theta} = \frac{n_{11}n_{22}}{n_{21}n_{12}}$$

The variance of $\ln(\hat{\theta})$ can be estimated as:

$$\hat{V} \left\{ \ln \left(\hat{\theta} \right) \right\} = \frac{1}{n_{11}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} + \frac{1}{n_{22}}$$

In-Class Problem 4

Leg Healing Trial Callum et al. 1992				
Leg Wound Healing				
	Healed	Not Healed	Total	P_{i+}
Elastic	35	30	65	0.538
Inelastic	19	48	67	0.284

Please calculate the relative risk and odds ratio of healing comparing elastic to inelastic bandages.

Please calculate the variance of the odds ratio estimate.

In-Class Problem 4

$$\hat{\theta} = \frac{35 \times 48}{19 \times 30} = 2.9474$$

The variance is determined for $\ln(\hat{\theta}) = 1.0809$.

$$\hat{V} \left\{ \ln(\hat{\theta}) \right\} = \frac{1}{35} + \frac{1}{30} + \frac{1}{48} + \frac{1}{19} = 0.13537$$

In-Class Problem 4

Form the confidence interval:

1 $\sqrt{V \left\{ \ln \left(\hat{\theta} \right) \right\}} = 0.36793.$

2 $1.96 * 0.36793 = 0.72114.$

3 $1.0809 - 0.72114 = 0.359777398$ and
 $1.0809 + 0.72114 = 1.802048025.$ This is the 95% confidence interval on the logarithmic scale.

4 Exponentiate to get back to the scale of $\hat{\theta}$. Therefore,
 $(e^{0.359777398}, e^{1.802048025}) = (1.443, 6.062).$

R Code

We can do this work in R. Here I'm using the *epiR* package:

```
library(epiR)
```

```
bandage<-matrix(data=c(35,19,30,48),nrow=2)
```

```
bandage<-as.table(bandage)
```

```
epi.2by2(bandage)
```

Example 4

Which produces the following output:

```
> epi.2by2(bandage)
      Outcome +   Outcome -   Total   Inc risk *
Exposed +         35         30       65   53.85 (41.03 to 66.30)
Exposed -         19         48       67   28.36 (18.01 to 40.69)
Total            54         78      132   40.91 (32.43 to 49.80)
```

Point estimates and 95% CIs:

```
-----
Inc risk ratio           1.90 (1.22, 2.95)
Inc odds ratio           2.95 (1.43, 6.06)
Attrib risk in the exposed * 25.49 (9.26, 41.72)
Attrib fraction in the exposed (%) 47.33 (18.05, 66.15)
Attrib risk in the population * 12.55 (-1.12, 26.22)
Attrib fraction in the population (%) 30.68 (7.22, 48.21)
-----
```

Uncorrected chi2 test that OR = 1: chi2(1) = 8.866 Pr>chi2 = 0.003

Fisher exact test that OR = 1: Pr>chi2 = 0.004

wald confidence limits

CI: confidence interval

* Outcomes per 100 population units